

```
import zipfile
with zipfile.ZipFile('/content/DataLabelling.zip', 'r') as z:
    z.extractall('/content/')
```

```
!pip install ultralytics -q
!cd /content && python train.py
```

	Class	Images	Instances	Box(P	R	
	all	72	185	0.904	0.827	
Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df_l_loss	ser
64/100	1.81G	0.4599	0.64	0.4945	0.915	
	Class	Images	Instances	Box(P	R	
	all	72	185	0.912	0.847	
Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df_l_loss	ser
65/100	1.81G	0.4379	0.5708	0.4894	0.9116	
	Class	Images	Instances	Box(P	R	
	all	72	185	0.93	0.838	
Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df_l_loss	ser
66/100	1.81G	0.4427	0.582	0.4872	0.9162	

```
import shutil
shutil.make_archive('/content/runs_shelf_inventory', 'zip', '/content/
print("Done – download runs_shelf_inventory.zip from the file browser"
```

Done – download runs_shelf_inventory.zip from the file browser

```
!python validate.py
```

Ultralytics 8.4.41 🚀 Python-3.12.13 torch-2.10.0+cu128 CUDA:0 (Tesla T4, YOLOv8n-seg summary (fused): 86 layers, 3,259,039 parameters, 0 gradients

val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 1138.5±555.1 MB/s, siz

val: Scanning /content/DataLabelling/valid/labels.cache... 72 images, 0 b

requirements: Ultralytics requirement ['faster-coco-eval>=1.6.7'] not fou

Using Python 3.12.13 environment at: /usr

Resolved 2 packages in 299ms

Prepared 1 package in 63ms

Installed 1 package in 6ms

+ faster-coco-eval==1.7.2

requirements: AutoUpdate success ✅ 0.8s

WARNING ⚠️ **requirements:** Restart runtime or rerun command for updates to

	Class	Images	Instances	Box(P	R	m
	all	72	185	0.939	0.883	0
	chocolate	21	38	0.874	0.737	0
	granola	15	33	0.931	0.909	
	mouse	2	3	1	0.89	0
	pen	8	23	0.903	0.913	0
	waterbottle	29	88	0.985	0.966	0

Speed: 3.3ms preprocess, 6.4ms inference, 0.0ms loss, 8.3ms postprocess p

Saving /content/runs/segment/val/predictions.json...

Results saved to /content/runs/segment/val

Validation Results:

```

Box mAP@50: 0.945
Box mAP@50-95: 0.843
Mask mAP@50: 0.944
Mask mAP@50-95: 0.814
Precision: 0.939
Recall: 0.883

```

```
!python validate.py
```

```

4.41 🚀 Python-3.12.13 torch-2.10.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
mmmary (fused): 86 layers, 3,259,039 parameters, 0 gradients, 11.3 GFLOPs
e access ✅ (ping: 0.0±0.0 ms, read: 1275.8±370.4 MB/s, size: 30.0 KB)
/content/DataLabelling/test/labels... 60 images, 0 backgrounds, 0 corrupt
created: /content/DataLabelling/test/labels.cache

```

Class	Images	Instances	Box(P	R	mAP50	mAP50-95
all	60	178	0.888	0.868	0.913	0.784
chocolate	16	41	0.883	0.756	0.817	0.659
granola	12	18	0.975	0.944	0.992	0.817
mouse	2	2	0.621	1	0.995	0.945
pen	9	36	0.959	0.658	0.767	0.537
waterbottle	23	81	1	0.983	0.995	0.960

```

reprocess, 7.5ms inference, 0.0ms loss, 8.4ms postprocess per image
t/runs/segment/val-2/predictions.json...
to /content/runs/segment/val-2

```

```

results:
0.913
: 0.784
0.887
: 0.772
88
68

```

```

import os
from collections import defaultdict

classes = ['chocolate', 'granola', 'mouse', 'pen', 'waterbottle']
splits = ['train', 'valid', 'test']
base = '/content/DataLabelling'
counts = defaultdict(lambda: defaultdict(int))

for split in splits:
    for f in os.listdir(f'{base}/{split}/labels'):
        with open(f'{base}/{split}/labels/{f}') as lf:
            for line in lf:
                cls_id = int(line.split()[0])
                counts[classes[cls_id]][split] += 1

print(f'{"Class":<15} {"Train":>7} {"Val":>7} {"Test":>7} {"Total":>7}')
print('-' * 42)
for cls in classes:

```

```
t, v, te = counts[cls]['train'], counts[cls]['valid'], counts[cls]
print(f'{cls:<15} {t:>7} {v:>7} {te:>7} {t+v+te:>7}')
```

Class	Train	Val	Test	Total
chocolate	161	38	41	240
granola	163	33	18	214
mouse	30	3	2	35
pen	130	23	36	189
waterbottle	341	88	81	510

```
for split in splits:
    imgs = len(os.listdir(f'{base}/{split}/images'))
    print(f'{split}: {imgs} images')
```

```
train: 293 images
valid: 72 images
test: 60 images
```

```
import os
from collections import defaultdict

classes = ['chocolate', 'granola', 'mouse', 'pen', 'waterbottle']
splits = ['train', 'valid', 'test']
base = '/content/DataLabelling'
counts = defaultdict(lambda: defaultdict(int))

for split in splits:
    for f in os.listdir(f'{base}/{split}/labels'):
        with open(f'{base}/{split}/labels/{f}') as lf:
            seen = set()
            for line in lf:
                cls_id = int(line.split()[0])
                seen.add(cls_id)
            for cls_id in seen:
                counts[classes[cls_id]][split] += 1

print(f"'Class':<15} {'Train':>7} {'Val':>7} {'Test':>7} {'Tot
print('-' * 42)
for cls in classes:
    t, v, te = counts[cls]['train'], counts[cls]['valid'], coun
    print(f'{cls:<15} {t:>7} {v:>7} {te:>7} {t+v+te:>7}')
```

Class	Train	Val	Test	Total
chocolate	68	21	16	105

granola	87	15	12	114
mouse	18	2	2	22
pen	33	8	9	50
waterbottle	113	29	23	165